

With frequent power cuts and rising power tariffs, alternative energy sources help you stay connected. We show you how you could *make hay while the sun shines!*

Hooked on to the Sun

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What you'll need

Solar panel 3Wp
Battery 6V/4.5Ah
ON/OFF switch (2 nos.)
1N4007 diode
White LEDs
General purpose PCB
Soldering iron
Soldering alloy
Connecting wire

**Total approximate
cost of components:
Rs. 900**

Assembling your own solar-powered study lamp is easy. In fact, Ankit, a Class 8 student assembled one at our office! We provided him with the required components and gave him the wiring diagram. We told him all the information provided here, and left him on his own while our photographer captured each step as it progressed.

Step 1

Gathering all your components is essentially the first step. Once

you're ready to assemble your study light, gather all your components on a clean table without any clutter. Plug in the soldering iron till it heats up. If heating is slow, switch off any cooler or fan in use.

Step 2

The first step is to arrange the white LEDs in series. As you can see, the anode and cathode leads appear sequentially.

Step 3

Once you've connected two LEDs in series, repeat the stages to have a minimum of 10 LEDs. You can increase the

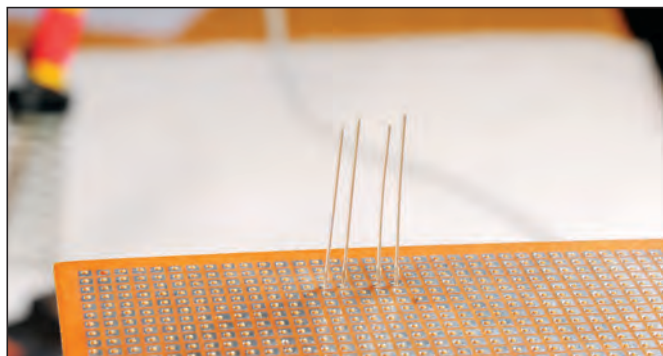
number of LEDs depending on the intensity you need. So products in the market have nearly 150 LEDs. However, this will increase the overall cost of your study lamp.

Step 4

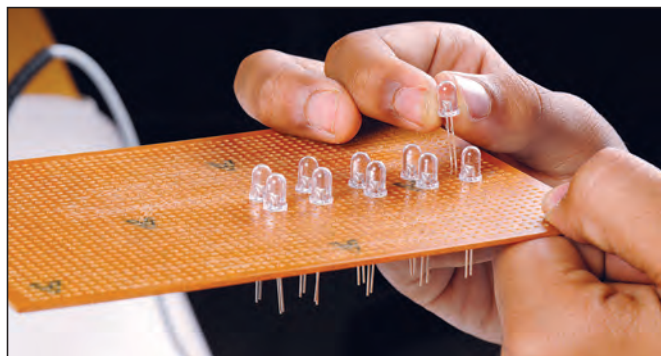
Once you have inserted the LEDs into the board, solder them. Being connected in parallel, the outer leads of the LEDs should be connected together.

Step 5

Cut the extra leads of the LEDs and connect the battery terminals to the LEDs using connecting wire.



Step 1: Start by arranging the LEDs in series



Step 2: Then you form a train of LEDs